

Retrieval Augmented Generation—Benefits and Concerns

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Abstract

Large Language Models (LLMs) have become increasingly popular since late 2022, but their limitations—especially hallucinations, static knowledge, and lack of domain-specific precision—have drawn growing concern. Retrieval-Augmented Generation (RAG) systems represent a promising solution by introducing an intermediary retrieval step that provides models with context-rich, up-to-date data. This literature review examines the key benefits and emerging concerns surrounding RAG-based LLMs, with a focus on the research question: What are the potential benefits and security implications of RAG systems? Research overwhelmingly agrees that RAG improves answer accuracy, relevance, and domain-specificity while also offering cost-saving, scalable architecture. However, opinions are more divided regarding security and privacy. While some scholars warn of vulnerabilities like poisoning attacks and data leakage, others argue that RAG can reduce reliance on sensitive training data, ultimately strengthening privacy protections. Both sides acknowledge that these risks can be mitigated through careful system design and data sanitization. The review concludes that RAG is not a flawless fix, but a compelling advancement that, if responsibly implemented, could significantly improve the usefulness and safety of LLMs. Future research should continue exploring secure integration practices, multi-modal retrieval, and more advanced synthesis of retrieved content to fully unlock the potential of RAG-enhanced AI systems.

Keywords: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Hallucination Mitigation, AI Security

Introduction

Large Language Models (LLMs) have exploded in popularity since the release of OpenAI's ChatGPT 3.5 in late 2022. They represent the most accessible avenue for collaboration between human and machine intelligence to date (Ghodratnama & Zakershaharak, 2024, p. 9). Unfortunately, due to the high costs of constantly training LLMs to keep up with current knowledge, as well as potentially disastrous hallucinations, they have difficulty “executing specialized, domain-specific tasks” (Ghodratnama & Zakershaharak, 2024, p. 3). In recent years, an exciting potential solution to both of these problems has been found: cutting-edge Retrieval-Augmented Generation (RAG) systems.

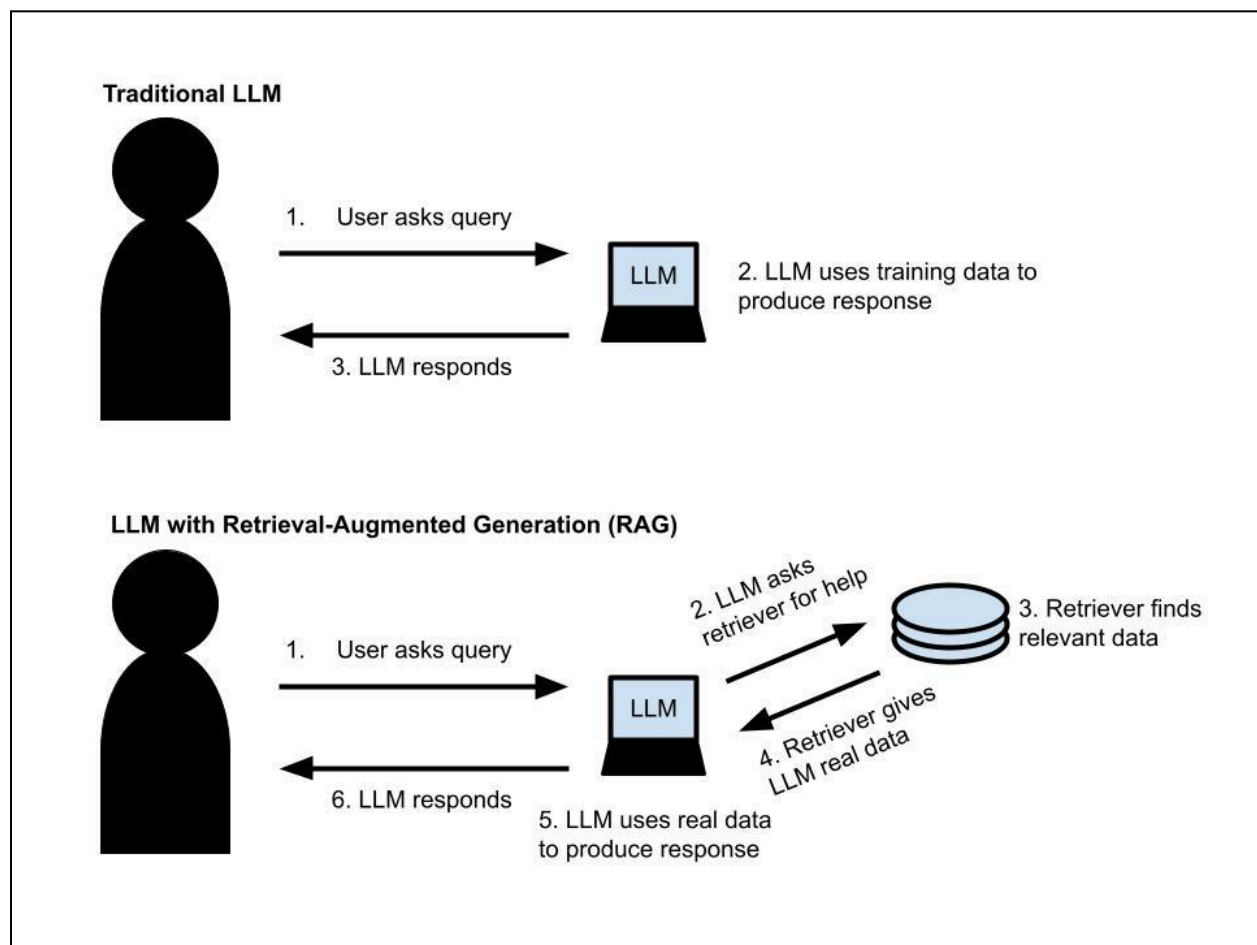


Figure 1. Diagram of an LLM with and without RAG integration (own image).

RAG systems introduce an intermediary data retrieval step into the LLM-querying process. Before sending a user's query to the LLM, the system locates relevant documents/data, which it then appends to the original query to provide richer context for the LLM to draw from in its answer. These retrieval-augmented queries decrease LLM hallucinations and improve answer accuracy and specificity (Fan et al., 2024; Huang et al., 2023; Sood et al., 2025) by providing LLMs with factual, ideally current, information. While these new systems fix some lingering issues, they also open up new questions for researchers.

The crux of the conversation about incorporating RAG systems into LLMs currently rests upon questions regarding the potential negative consequences RAG systems may introduce and how those consequences can be mitigated (He et al., 2025; Huang et al., 2023; Zeng et al., 2024). Research on new RAG-related security & privacy vulnerabilities is coming out rapidly, as well as research proposing methods to implement retrieval systems more securely. As innovators rush to improve their LLMs with RAG systems, it is vital that they have access to a current, digestible examination of RAG-related risks and recommended implementation practices to inform them. I aim to serve researchers by writing a literature review focusing on the research question "What are the potential benefits and security implications of RAG systems?"

My paper will have just two main body sections, which explore this subject deeply: 1. The benefits of integrating RAG systems with LLMs, and 2. Security & privacy implications they introduce. I will also briefly outline future research topics mentioned in the literature.

The Benefits of RAG-Based LLMs

Researchers seem to almost unanimously agree that RAG integration can provide significant improvements to LLMs. In the literature, there were three main trends in the kinds of

benefits discussed: 1. Improved quality of information, 2. Increased specialization opportunities, or “domain-specificity”, and 3. Benefits in enterprise/business.

Quality of Information

Most sources agree that one of the principal benefits of integrating RAG into LLMs is improved quality of information utilized and returned by the LLM. (Fan et al., 2024; Filipovska et al., 2025; Ghodratnama & Zakershahra, 2024; Gokcimen & Das, 2025; Han et al., 2024, Huang et al., 2023; Sood et al., 2025, Xiong et al., 2024) The literature focuses on three major ways this manifests: 1. Hallucination mitigation, 2. Specificity of responses, and 3. Currency of information.

A detrimental and unfortunately common problem with LLMs has been “hallucination”, or the generation of “plausible sounding but factually incorrect responses” (Han et al., 2024, p. 1). These responses significantly detract from the trustworthiness of language models, especially in the context of highly sensitive fields like scholarship, law, and medicine. In a study specifically focused on hallucination in AI systems, Sood et al. (2025) reported that RAG methods help “mitigate hallucinations in AI-generated responses by grounding the model’s output in factual, retrieved knowledge rather than relying solely on its pre-trained parameters” (p. 10). Han et al. (2024) echo the benefits of this “grounding” of model output: “RAG-based LLMs have significantly improved accuracy, relevance, and contextual comprehension” (p. 14). This increase of accuracy and relevance not only improves LLM users’ experiences by mitigating hallucinations, but it also improves the quality of the answers they receive through greater specificity.

LLMs are limited in the specificity of their answers by the generalized nature of their training data. While this generalized training makes an LLM a good “chatbot”, more refined data

is required to leverage an LLM as a true “assistant”, which “differs from an LLM-based chatbot in its purpose and behavior.” (Alier et al., 2025, p. 3). Alier et al. (2025) further report that integrating RAG into an LLM allows it to “[utilize] reliable sources of information, which it can accurately cite and link while maintaining the natural language processing, reasoning, and instruction-following capabilities of an LLM” (p. 3). Other sources specify further that having access to current information is a major boon to RAG-based LLMs, as opposed to LLMs without RAG, which rely completely on “static” knowledge (Han et al., 2024, p. 1). Xiong et al. (2024) outline the benefits of connecting LLMs to search engines through RAG systems, stating that “allowing [LLMs] to access and incorporate real-time data and diverse content from the web... [enables] them to provide more accurate, contextually relevant, and up-to-date responses to user queries” (p. 1). These benefits are being sought out by companies, with Fan et al. (2024) pointing out that “recent efforts have been made to take advantage of RAG to enhance the capabilities of LLMs in various tasks, especially those [with high demand] for the latest and reliable knowledge” (p. 2). The specificity and timeliness of the information available to LLMs through RAG may provide powerful opportunities in even specialized industries.

Domain Specificity

Many sources agree that having access to reliable, specific, and current sources of information through RAG is a key way to empower LLMs in “domain-specific” tasks (Alier et al., 2025; Fan et al., 2024; Filipovska et al., 2025; Han et al., 2024; He et al., 2025; Rathod et al., 2025; Sood et al., 2025; Xiong et al., 2024). Recent studies have explored specializing LLMs through RAG in various fields. Alier et al. (2025) created a framework for implementing RAG-based LLMs into education through their modular “Learning Assistant Manager and Builder”, or LAMB. They reported satisfaction with the results of their RAG integration, which

combined LLMs' strengths with "specific directions and knowledge retrieval that can be referenced and validated, [and] tailored for specific learning and teaching situations safely and ethically" (p. 11). In contrast to Alier et al.'s focus on how RAG-based LLMs can improve classroom learning, Han et al. (2024) studied how researchers can leverage RAG integration and LLMs to automate systemic literature reviews. They concluded that RAG-based LLMs are "promising candidates" for "automating...data extraction, summarization, research synthesis, and trend identification" (p. 14). Multiple studies discussed the promising potential use of RAG to enhance LLMs in medical fields (Rathod et al., 2025; Huang et al., 2023). Huang et al. (2023) particularly mentioned RAG's use in many areas: "Currently, RAG is widely used in various scenarios, including disease identification, medication counseling, search engines, customer service, and email/code completion, providing substantial benefits to multiple industries" (p. 1). Not only can RAG empower LLMs to perform as domain-specific assistants, but the inherent architecture of RAG systems provides financial and efficiency-related benefits to those who implement it.

Benefits in Business

Aside from RAG improving the abilities of LLMs, the innate architecture of RAG carries benefits of 1. Flexibility, 2. Scalability, and 3. Cost-saving. Zeng et al. (2024) discuss how RAG architecture improves LLM flexibility: "One of RAG's distinctive features is its flexible architecture, allowing for the seamless interchange or update of its three core components: the dataset, the retriever, and the LLM" (p. 2). They continue, pointing out that the separation of these components means that "adjustments to any of these elements can be made without necessitating re-training or fine-tuning of the entire system." This flexibility makes modular

LLM tools, like the previously mentioned Learning Assistant Manager and Builder of Alier et al. (2025), possible. Using RAG architecture, companies creating LLM tools can focus on creating specialized databases and retrievers while allowing their LLMs to be more general. The scalability and cost-saving benefits of RAG architecture both have to do with the high training costs of LLMs. LLMs are notoriously expensive to train, which may be part of the reason why large tech companies have created many of the most dominant models (OpenAI's ChatGPT, Google's Gemini, and Meta's LLaMA, for instance). Ghodratnama & Zakershahra (2024) clearly state how companies save resources when using RAG architecture rather than relying on fine-tuning of a model: "The potential for cost and computational savings is evident, as the need for frequent, exhaustive retraining of the model is mitigated, supplanted by updates to the retrieval database" (p. 20). Implementing RAG architecture can benefit companies with fewer training resources, as RAG integrations can improve lower-cost LLMs by "expanding their knowledge base beyond pre-training/fine-tuning datasets and enabling them to provide more accurate, contextually relevant, and up-to-date responses to user queries" (Xiong et al., 2024).

In summary, RAG systems are generally believed to present potential benefits to general LLM-using consumers, companies, and industries at large. RAGs partially address some of the most prevalent concerns with the adoption of LLMs into industries, including response reliability/hallucination mitigation, lack of domain-specific knowledge, and high training/fine-tuning costs.

Security & Privacy Implications

LLMs have been targeted for various forms of exploitation since they increased in popularity, but the stakes of such exploitation have generally been quite low. The most common method of exploitation has been "prompt injection", where LLM users tailor their prompt to

manipulate the LLM into saying something opposed to its training, or to reveal its training data. The worst consequences of most of these exploitation attempts seem to have simply been some embarrassment for the company in charge of the LLM. However, consequences of LLM exploitation may increase in severity when they are given access to specific, perhaps sensitive, data through RAG architecture. Unlike the broad acknowledgement of the benefits of RAG-based LLMs, the opinions of researchers are much more varied on the possible security and privacy implications of integrating RAG with LLMs.

RAG and Security

Though some researchers are especially cautious, opinions about the effect of RAG on the security of LLMs, and cybersecurity in general, are fairly positive. Fan et al. (2024) caution readers, saying “Recent research indicates that RA-LLMs can be maliciously and unintentionally manipulated to make unreliable decisions and harm humans, which may have serious consequences in safety-critical scenarios” (p. 13). In a similar vein, Das et al. (2025) point out that the added architecture of databases & retrievers to LLMs introduces new possible targets of “poisoning attacks”, where malicious attackers “[inject] a small number of poisoned samples into the RAG knowledge base” (p. 18). While the databases & retrievers underlying the RAG architecture may be vulnerable to new methods of attack, some researchers also discuss how they can actually contribute to the security of the LLM.

RAG integration can have positive impacts on the security of proprietary LLM data, as well as how LLMs interact with cybersecurity. As mentioned previously, one motive behind prompt injection attacks is to see the prompts & underlying training data of an LLM. Rathod et al. (2025) explain how RAG integration can protect LLMs from these attacks: “RAG integrates LLMs with external knowledge bases to minimize the use of sensitive training data...reduc[ing]

the model’s dependency on memorized content” (p. 750). Zeng et al. (2024) agree, saying “integrating retrieval data reduces the risk of privacy leaks from LLMs’ training data, making it harder for attackers to access this information” (p. 8). Even beyond protecting the LLM system that it is directly integrated with, Soot et al. (2025) found that the hallucination-mitigation effect of RAG integration had an impressive effect on the Artificial Intelligence-enhanced cybersecurity systems they studied:

By implementing RAG...AI-driven, threat-hunting systems can generate more accurate insights based on live security data instead of relying solely on static training...Combining RAG for grounded knowledge retrieval and [uncertainty estimation techniques] creates an accurate and trustworthy AI-driven security framework. This significantly reduces the risk of hallucinated threat reports in real-world cybersecurity environments. (Sood et al., 2025, pp. 10–12)

In summary, RAG architecture can be a boon to the security of LLM systems, while introducing a few more vulnerable points in their database and retrieval components.

RAG and Privacy

Similarly to discussion on the security implications of RAG systems, discussion on their privacy implications has also been divided. Interestingly, opinions in this area seem to be especially polarized; there appear to be two main schools of thought on the matter: 1. RAG-based LLMs have an increased risk of leaking private data (Fan et al., 2024; He et al., 2025; Huang et al., 2023), and 2. RAG-based LLMs have a decreased risk of leaking private data (Ghodratnama & Zakershahrak, 2024; Gokcimen & Das, 2025; Rathod et al., 2025; Zeng et al., 2024).

Most of the “increased risk”-believing group seem to be concerned about the theoretical, future possibility of data leakage. After bringing up security concerns, as previously mentioned, Fan et al. (2024) continue, saying “In addition, private retrieval database[s] [have] a risk of leakage, raising concerns regarding the privacy of RA-LLMs” (p. 13). He et al. (2025) agree, stating “RAG systems could encounter substantial privacy challenges when private data is involved in the retrieval process” (p. 1). In contrast to the future-based concerns of Fan et al. (2024) and He et al. (2025), Huang et al. (2023) conducted a numbers-backed study of the privacy implications of RAG-based LLMs, finding that “Incorporating a private datastore in [LLMs] leads to increased privacy risks (both targeted and untargeted) compared to [traditional] language models trained on private data” (p. 8). In sum, whether in the present or in the future, this group is concerned about the integration of private data into LLMs.

On the other hand, the “decreased risk”-believing group seem to think that RAG systems can be used with enough finesse to prevent data leakage. Ghodratnama & Zakershahra (2024) report that RAG databases are more secure than LLM training data, saying “the approach of RAG also lends a hand in addressing privacy concerns. Since RAG pulls from an external retrieval database, instead of integrating data into the model, it aids in keeping sensitive information more secure” (p. 20). Some researchers further emphasize that RAG architecture provides enhanced security inherently, with Gokcimen & Das (2025) saying “RAG architecture prevents data leakage by conducting the entire process in a secure environment” (p. 77). Using medical-focused LLMs as a case study, Rathod et al. (2025) explain:

By incorporating RAG, healthcare LLMs can retrieve only the necessary, non-sensitive information from external databases, ensuring privacy while answering queries...RAG integrates LLMs with external knowledge bases to minimize the use of sensitive training

data. By retrieving specific, domain-relevant information at runtime, RAG reduces the model's dependency on memorized content, mitigating data leakage risks. Particularly effective in healthcare for retrieving anonymized medical guidelines. (pp. 748–750)

Zeng et al. (2024) agree, with the small caveat that private data should be processed and anonymized during retrieval: “RAG can effectively protect private information in LLMs’ training data. Using non-sensitive public or carefully desensitized data as retrieval content can greatly minimize the risk of information leakage from LLMs” (p. 8). This caveat presents a point on which these two polarized groups agree.

Risk Mitigation

Both the “increased risk”- and “decreased risk”-believing groups agree that RAG integration into LLMs is a worthy pursuit, and they agree that steps can be taken to further secure RAG architecture. Most of the research into risk-mitigation strategies is being done by the “increased risk”-believing group. Fan et al. (2024) say “developing trustworthy RA-LLMs is of paramount importance” (p. 13), with He et al. (2025) echoing that importance, saying “designing a privacy-preserving RAG system is of great practical significance and critical urgency” (p. 2). Various groups are approaching this research into risk-mitigation strategies in RAG systems from different angles. Fan et al. (2024) outlined some general goals for future RAG-based LLM innovators to aspire to: “1) robustness, 2) fairness, 3) explainability, and 4) privacy” (p. 13). Gokcimen & Das (2025) created an intricate multi-layered framework that protects LLMs from hallucination and keeps their retrieval and database components safe. He et al. (2025) have critically rejected some current privacy-preserving approaches due to “1. Lack of Rigorous Theoretical Guarantees...2. Potential Risk of Privacy Leakage... [and] 3. Excessive Depletion of Valid Information” (p. 2), and demonstrated a novel method of data sanitization which is more

rigorous. Their findings are in line with the recommendations of Huang et al. (2023), who studied various methods of data sanitization and proposed a very similar idea. In summary, all of these studies have been complementary and are demonstrating more secure ways of integrating RAG architecture into LLMs.

To conclude this section, the discussions around security implications of RAG, privacy concerns, and the possible mitigation of each have been polarizing, yet thorough and fruitful.

Future Areas of Research

Briefly, there are two main areas of future research put forth by the literature. The first, as just discussed, is how the security & privacy of RAG systems can be improved. The second is possible future applications of RAG systems. Some proposed applications include multi-lingual enhancement of LLMs through RAG (Fan et al., 2024), RAG systems that can retrieve and utilize multi-modal data (Fan et al., 2024), improvement of RAG-based LLMs' abilities to synthesize multiple retrieved sources in answering questions (Han et al., 2024), and how to best integrate RAG with various kinds of databases (Alier et al., 2025).

Conclusion

RAG systems appear to be a promising solution to several of the limitations of traditional LLMs. As the literature overwhelmingly agrees, integrating retrieval into the LLM query process can significantly improve the quality, specificity, and timeliness of outputs. This grounding effect not only mitigates hallucinations but allows for more specific answers, particularly in domain-specific scenarios. RAG architecture opens doors for LLMs to function less like generalized chatbots and more like knowledgeable, task-focused assistants. On top of these functional improvements, RAG also provides flexibility and scalability that can help reduce the cost of utilizing LLMs, which especially benefits smaller companies and research institutions.

At the same time, RAG integration raises serious questions about security and privacy. In contrast to the near-unanimous support for RAG's benefits, opinions on these risks are more divided. Some researchers warn of increased vulnerability due to the new vulnerabilities introduced by retrievers and external databases. Others argue that RAG can actually reduce privacy concerns by limiting the reliance on sensitive pre-training data. Either way, both sides agree that risk mitigation is both possible and necessary. Encouragingly, current work is already proposing and testing ways to safeguard RAG-based LLMs, such as better data sanitization methods and multi-layered security frameworks.

In short, the addition of RAG to LLMs represents a significant step forward. It offers meaningful solutions to current problems while also requiring thoughtful implementation to avoid introducing new ones. As researchers continue to refine RAG systems, future studies will be critical in answering still-open questions—how to best anonymize retrieved data, how to handle multi-modal input/output, and how to synthesize information from multiple sources effectively. Addressing these challenges will make RAG systems safer, more powerful, trustworthy, and adaptable in the expansion of LLMs usage in our world.

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